

OS MSE

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Q: Operating Systems: Overview

Hinglish Note:

Operating System ek software hota hai jo computer ke hardware aur user ke beech bridge ka kaam karta hai. Ye memory manage karta hai, files handle karta hai, devices ko control karta hai aur programs ko smoothly run karne me help karta hai. Basically OS hi computer ko usable banata hai.

English Note:

An Operating System is a software layer that works as a bridge between the computer hardware and the user. It manages memory, files, devices, and ensures programs run smoothly. In simple words, the OS makes the computer usable.

Q: OS Structure

Hinglish Note:

OS structure batata hai ki Operating System andar se kaise organized hai. Isme layers hoti hain jaise hardware, kernel, system calls aur user applications. Kernel sabse important part hota hai jo hardware ko direct control karta hai. Layers se system simple, secure aur manageable ban jata hai.

English Note:

The OS structure shows how an Operating System is organized internally. It has layers like hardware, kernel, system calls, and user applications. The kernel is the core part that controls the hardware directly. These layers make the system simple, secure, and easier to manage.

Q: OS Services

Hinglish Note:

OS services woh functions hain jo Operating System user aur programs ko provide karta hai. Jaise program execution, file management, memory management,

device control, security aur communication. Ye services system ko smooth aur user-friendly banati hain.

English Note:

OS services are the functions provided by the Operating System to users and programs. These include program execution, file management, memory management, device control, security, and communication. These services keep the system smooth and user-friendly.

Q: System Calls

Hinglish Note:

System calls wo special requests hoti hain jo program OS se karta hai kuch kaam karne ke liye, jaise file read/write, process create, memory allocate. Ye user programs aur OS ke beech ek interface ka kaam karti hain. Direct hardware access nahi hota, sab system calls ke through hota hai.

English Note:

System calls are special requests made by a program to the Operating System to perform tasks like reading or writing files, creating processes, or allocating memory. They act as an interface between user programs and the OS. Direct hardware access is not allowed, everything goes through system calls.

Q: Types of OS

Hinglish Note:

Operating Systems ke alag-alag types hote hain jo use-case ke hisaab se design kiye gaye hote hain. Common types me batch OS, multiprogramming OS, time-sharing OS, real-time OS, distributed OS, parallel OS aur embedded OS aate hain. Har type ka apna style hota hai tasks ko manage karne ka.

English Note:

There are different types of Operating Systems designed for different purposes. Common types include batch OS, multiprogramming OS, time-sharing OS, real-time OS, distributed OS, parallel OS, and embedded OS. Each type has its own way of managing tasks efficiently.

Q: Batch Operating System

Hinglish Note:

Batch OS me tasks ko groups ya batches me collect karke ek sath process kiya jata hai. User direct system se interact nahi karta. Yeh system tab useful hota hai jab similar type ke jobs ko automate karke run karna ho.

English Note:

In a Batch Operating System, tasks are collected into groups or batches and processed together. The user does not interact directly with the system. This type of OS is useful when similar jobs need to be automated and executed in bulk.

Q: Multi-programmed Operating System**Hinglish Note:**

Multi-programmed OS me ek time par multiple programs memory me loaded hote hain. CPU ek program ke wait time me dusra program process kar leta hai. Isse CPU utilization badh jata hai aur system fast lagta hai.

English Note:

A multi-programmed OS keeps multiple programs loaded in memory at the same time. When one program is waiting, the CPU switches to another program. This increases CPU utilization and makes the system feel faster.

Q: Time Sharing Operating System**Hinglish Note:**

Time-sharing OS me multiple users ek hi system ko same time par use kar sakte hain. CPU har user ko thoda-thoda time slot deta hai. Isse sabko lagta hai ki system unke liye hi kaam kar raha hai. Ye fast response aur multitasking ke liye perfect hai.

English Note:

A time-sharing OS allows multiple users to use the same system at the same time. The CPU gives each user a small time slot. This makes everyone feel like the system is working only for them. It's ideal for fast response and multitasking.

Q: Real-time Operating System

Hinglish Note:

Real-time OS wo hota hai jo tasks ko exact time limit me complete karta hai. Isme delay allowed nahi hota. Ye systems critical areas me use hote hain jaise medical devices, robots, flight control systems. Timing sabse important factor hota hai.

English Note:

A real-time OS completes tasks within a strict time limit. Delays are not acceptable. These systems are used in critical areas like medical devices, robotics, and flight control systems. Timing is the most important factor here.

Q: Distributed Operating System

Hinglish Note:

Distributed OS multiple computers ko ek network me jod kar unhe ek single system ki tarah behave karate hain. Users ko lagta hai ki sab resources ek hi machine par available hain. Isse speed, reliability aur resource sharing improve hoti hai.

English Note:

A distributed OS connects multiple computers in a network and makes them work like a single system. Users feel as if all resources are available on one machine. This improves speed, reliability, and resource sharing.

Q: Parallel Operating System

Hinglish Note:

Parallel OS me multiple processors ek sath kaam karte hain taaki tasks fast complete ho sake. Ek bada task ko chhote parts me tod kar alag-alag processors par run karaya jata hai. Isse speed aur performance dono improve hote hain.

English Note:

In a Parallel OS, multiple processors work together to complete tasks faster. A large task is divided into smaller parts and executed across different processors.

This increases both speed and overall performance.

Q: Embedded Systems

Hinglish Note:

Embedded systems chhote, special-purpose systems hote hain jo kisi device ke andar specific kaam karte hain. Jaise washing machine controllers, microwave, car systems. Ye fast, reliable aur limited resources me kaam karne ke liye design hote hain.

English Note:

Embedded systems are small, special-purpose systems that perform specific tasks inside a device. Examples include washing machine controllers, microwaves, and car control units. They are designed to be fast, reliable, and work within limited resources.